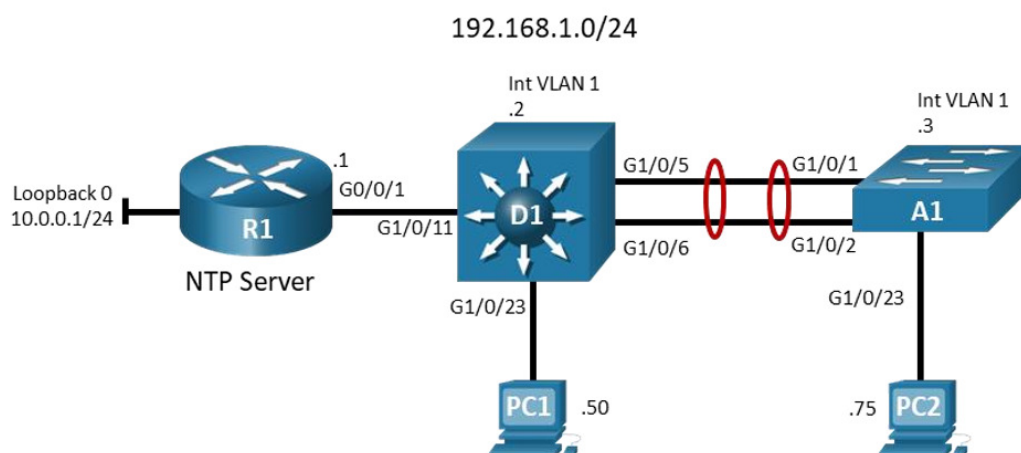




Lab - Implement Flexible Netflow

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/1	192.168.1.1	255.255.255.0	N/A
	Loopback0	10.0.0.1	255.255.255.0	N/A
D1	VLAN 1	192.168.1.2	255.255.255.0	N/A
A1	VLAN 1	192.168.1.3	255.255.255.0	N/A
PC1	NIC	192.168.1.50	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.75	255.255.255.0	192.168.1.1

Objectives

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

Part 2: Configure and Verify Flexible Netflow

Part 3: (Optional) Explore Netflow Collector and Analyzer Software

Background / Scenario

NetFlow is a Cisco IOS technology that provides statistics on packets flowing through the router. NetFlow is the standard for acquiring IP operational data from IP networks. NetFlow provides data to enable network and security monitoring, network planning, traffic analysis, and IP accounting.

Flexible NetFlow improves on original classic NetFlow by adding the capability to customize the traffic analysis parameters for your specific requirements. Flexible NetFlow facilitates the creation of more complex configurations for traffic analysis and data export through the use of reusable configuration components.

Note: This lab is an exercise in configuring options available for Flexible Netflow and does not necessarily reflect network troubleshooting best practices.

Note: Make sure that the switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 1 Router (Cisco 8200 with Cisco IOS XE Release 17.9.5a universal image or comparable)
- 1 Switch (Cisco 9300L with Cisco IOS XE Release 17.9.5a universal image or comparable)
- 1 Switch (Cisco 9200L with Cisco IOS Release 17.9.5a universal image or comparable)
- 2 PCs (Choice of operating system with terminal emulation program and a packet capture utility installed, such as Wireshark)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on routers.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each device.

- a. Console into each device, enter global configuration mode, and apply the basic settings. The startup configurations for each device are provided below.

Router R1

```
hostname R1
no ip domain lookup
line con 0
  exec-timeout 0 0
  logging synchronous
exit
```

```
line vty 0 4
  privilege level 15
  exec-timeout 0 0
  password cisco123
  login
  transport telnet
  exit
interface g0/0/1
  ip address 192.168.1.1 255.255.255.0
  no shutdown
  exit
interface loopback 0
  ip address 10.0.0.1 255.255.255.0
  no shutdown
  exit
ntp master 3
end
```

Switch D1

```
hostname D1
no ip domain lookup
line con 0
  exec-timeout 0 0
  logging synchronous
  exit
line vty 0 4
  privilege level 15
  exec-timeout 0 0
  password cisco123
  login
  exit
interface vlan 1
  ip address 192.168.1.2 255.255.255.0
  no shutdown
  exit
ip default-gateway 192.168.1.1
interface g1/0/23
  spanning-tree portfast
  switchport mode access
  no shutdown
  exit
interface g1/0/11
  spanning-tree portfast
  switchport mode access
  no shutdown
  exit
interface range g1/0/5-6
  switchport mode trunk
  channel-group 1 mode active
  no shutdown
  exit
interface range g1/0/1-4, g1/0/7-10, g1/0/12-22, g1/0/24, g1/1/1-4
  shutdown
  exit
ntp server 192.168.1.1
end
```

Switch A1

```
hostname A1
no ip domain lookup
line con 0
  exec-timeout 0 0
  logging synchronous
  exit
line vty 0 4
  privilege level 15
  exec-timeout 0 0
  password cisco123
  login
  exit
interface vlan 1
  ip address 192.168.1.3 255.255.255.0
  no shutdown
  exit
ip default-gateway 192.168.1.1
interface range g1/0/1-2
  switchport mode trunk
  channel-group 1 mode active
  no shutdown
  exit
interface g1/0/23
  switchport mode access
  spanning-tree portfast
  no shutdown
  exit
interface range g1/0/3-22, g1/0/24
  shutdown
  exit
ntp server 192.168.1.1
end
```

- b. Set the clock on each device to UTC time.
- c. Configure IPv4 addresses on hosts PC1 and PC2 as shown in the addressing table.
- d. Verify that R1, D1, A1, and PC2 can successfully ping PC1 at 192.168.1.50.

Part 2: Configure and Verify Flexible Netflow

As previously stated, Flexible Netflow provides the ability to customize traffic analysis parameters. The workflow for Flexible Netflow consists of four steps:

- Step 1.** Create Flow Records. Flow records define the information to be collected. There are predefined flow records that match the flow caching done by Classic Netflow, or you can configure your own custom flow record to suit your needs.
- Step 2.** Create Flow Exporter. This defines where compiled statistic information is sent.
- Step 3.** Create Flow Monitor and associate Flow Records and Flow Exporters with it.
- Step 4.** Configure the appropriate interface for input or output caching associated with the appropriate Flow Monitor.

In this part of the lab, you will configure Flexible Netflow to send statistical information about R1 interface g0/0/1 to PC1.

Step 1: Create flow records.

- a. For our first flow record, we will use the predefined ipv4 original-input flow record. Because it is predefined, there is no configuration necessary.
- b. For our second flow record, we will create a custom flow record. Because the first flow record is focused on input traffic, the second will focus on output traffic. Create a flow record named CCNP8-CUSTOMOUT.

```
R1(config)# flow record CCNP8-CUSTOM-OUT
```

- 1) Give the flow record a description.

```
R1(config-flow-record)# description Custom Flow Record for outbound traffic 2)
```

Set up the flow record to match ipv4 destination address and transport destination.

```
R1(config-flow-record)# match ipv4 destination address
R1(config-flow-record)# match transport destination-port
```

- 3) Set up the flow record to collect bytes and packets.

```
R1(config-flow-record)# collect counter bytes
R1(config-flow-record)# collect counter packets
```

- 4) Use the **show flow record CCNP8-CUSTOM-OUT** command to examine the results.

```
R1# show flow record CCNP8-CUSTOM-OUT
flow record CCNP8-CUSTOM-OUT:
  Description:          Custom Flow Record for outbound traffic
  No. of users:         0
  Total field space:    14 bytes
  Fields:
    match ipv4 destination address
    match transport destination-port
    collect counter bytes
    collect counter packets
```

Step 2: Create a flow exporter.

- a. The flow exporter configuration defines where the cached information will be sent. Create a flow exporter named CCNP8-COLLECTOR-HOST. Further specify that the exporter should use Netflow version 9, and point to 192.168.1.50 udp port 9999.

```
R1(config)# flow exporter CCNP8-COLLECTOR-HOST
R1(config-flow-exporter)# destination 192.168.1.50
R1(config-flow-exporter)# export-protocol netflow-v9
R1(config-flow-exporter)# transport UDP 9999
R1(config-flow-exporter)# exit
```

- b. Use the **show flow exporter CCNP8-COLLECTOR-HOST** command to examine the results.

```
R1# show flow exporter CCNP8-COLLECTOR-HOST
Flow Exporter CCNP8-COLLECTOR-HOST:
  Description:          User defined
  Export protocol:      NetFlow Version 9
  Transport Configuration:
    Destination IP address: 192.168.1.50
    Source IP address:    192.168.1.1
```

```
Transport Protocol:    UDP
Destination Port:     9999
Source Port:          63275
DSCP:                 0x0
TTL:                  255
Output Features:      Used
```

Step 3: Create flow monitors.

The flow monitor associates a flow record with the flow exporter. For our exercise, we need to create two flow monitors, one for each flow record.

- a. Create the first flow monitor and name it CCNP8-INBOUND-MONITOR using the **flow monitor CCNP8-INBOUND-MONITOR** command. As part of the flow monitor, specify that it will record the netflow ipv4 original-input flow record, export the cache to the exporter every 30 seconds, and identify CCNP8COLLECTOR-HOST as the exporter.

```
R1(config)# flow monitor CCNP8-INBOUND-MONITOR
R1(config-flow-monitor)# record netflow ipv4 original-input
R1(config-flow-monitor)# cache timeout active 30
R1(config-flow-monitor)# exporter CCNP8-COLLECTOR-HOST
R1(config-flow-monitor)# exit
```

- b. Create the second flow monitor and name it CCNP8-OUTBOUND-MONITOR using the **flow monitor CCNP8-OUTBOUND-MONITOR** command. As part of the flow monitor, specify that it will record the CCNP8-CUSTOM-OUT flow record, export the cache to the exporter every 30 seconds, and identify CCNP8-COLLECTOR-HOST as the exporter.

```
R1(config)# flow monitor CCNP8-OUTBOUND-MONITOR
R1(config-flow-monitor)# record CCNP8-CUSTOM-OUT
R1(config-flow-monitor)# cache timeout active 30
R1(config-flow-monitor)# exporter CCNP8-COLLECTOR-HOST
R1(config-flow-monitor)# exit
```

- c. Use the **show flow monitor** command to examine the results.

```
R1# show flow monitor
Flow Monitor CCNP8-INBOUND-MONITOR:
  Description:      User defined
  Flow Record:      netflow ipv4 original-input
  Flow Exporter:    CCNP8-COLLECTOR-HOST
  Cache:
    Type:           normal (Platform cache)
    Status:         not allocated
    Size:           200000 entries
    Inactive Timeout: 15 secs
    Active Timeout:  30 secs
    Trans end aging: off
```

```
Flow Monitor CCNP8-OUTBOUND-MONITOR:
Description:      User defined
Flow Record:      CCNP8-CUSTOM-OUT
Flow Exporter:    CCNP8-COLLECTOR-HOST
Cache:
  Type:           normal (Platform cache)
  Status:         not allocated
  Size:           200000 entries
  Inactive Timeout: 15 secs
  Active Timeout:  30 secs
  Trans end aging: off
```

Step 4: Configure the interface for flow caching.

The last step is to configure the appropriate interface(s) so that they will cache information. In our lab, we will focus on the input and output from interface g0/0/0 on R1. Use the **ip flow monitor <name> <direction>** command on g0/0/1 to specify the inbound and outbound flow monitors you have created.

```
R1(config)# interface g0/0/1
R1(config-if)# ip flow monitor CCNP8-INBOUND-MONITOR input
R1(config-if)# ip flow monitor CCNP8-OUTBOUND-MONITOR output
R1(config-if)# exit
```

Step 5: Create some traffic.

To gather statistics, we will need some traffic.

- From PC2, start a continuous ping to R1. As a part of each set of ping parameters, set the size of the packets to 1475 bytes.

The windows commands are as follows:

```
C:\> ping 10.0.0.1 -t -l 1475
```

- From switch A1, telnet to R1. Login and leave the session running.
- From switch D1, use the extended ping utility to send pings to R1 Loopback 0 using a sweep range of 36 bytes to 18024 bytes. Set the repeat count to 1,000,000 and the sweep interval to 1.

```
D1# ping
Protocol [ip]:
Target IP address: 10.0.0.1
Repeat count [5]: 100000
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Ingress ping [n]:
Source address or interface:
DSCP Value [0]:
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0x0000ABCD]:
```

```
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]: y
Sweep min size [36]:
Sweep max size [18024]:
Sweep interval [1]:
Type escape sequence to abort.
Sending 89945, [36..18024]-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

Note: escape sequence Ctrl-Shift-6

- d. On PC1, start Wireshark and apply the filter **ip.src == 192.168.1.1 && udp.dstport == 9999 && ! icmp**. This filters the display to show packets from 192.168.1.1 destined for UDP port 9999 and not ICMP packets.

Step 6: Wait 60 seconds then examine the results.

- a. On PC1, observe the Wireshark display window. There should be traffic matching the filter being displayed.
- b. On R1, issue the command **show flow monitor CCNP8-INBOUND-MONITOR statistics**.

```
R1# show flow monitor CCNP8-INBOUND-MONITOR statistics
Cache type:                               Normal (Platform cache)
Cache size:                               200000
Current entries:                           2
High Watermark:                           12
Flows added:                               103
Flows aged:                                101
- Active timeout      (   30 secs)         38
- Inactive timeout    (   15 secs)         63
```

- c. On R1, issue the command **show flow monitor CCNP8-INBOUND-MONITOR cache**. Note: Output will vary depending upon how long within the 30-second window traffic has been caching.

```
R1# show flow monitor CCNP8-INBOUND-MONITOR cache
Cache type:                               Normal (Platform cache)
Cache size:                               200000
Current entries:                           1
High Watermark:                           12
Flows added:                               112
Flows aged:                                111
- Active timeout      (   30 secs)         43
- Inactive timeout    (   15 secs)         68
```

```
IPV4 SOURCE ADDRESS:    192.168.1.75
IPV4 DESTINATION ADDRESS: 10.0.0.1
TRNS SOURCE PORT:       0
TRNS DESTINATION PORT:  2048
INTERFACE INPUT:        Gi0/0/1
FLOW SAMPLER ID:         0
IP TOS:                  0x00 IP
PROTOCOL:                1
```



```
ip source as:          0
ip destination as:     0
ipv4 next hop address: 0.0.0.0
ipv4 source mask:      /0
ipv4 destination mask: /0
tcp flags:             0x00
interface output:      Null counter
bytes:                 12024 counter
packets:               8
timestamp first:       20:43:34.189
timestamp last:        20:43:41.263
```

d. Stop all the pings and exit the telnet session.

Part 3: (Optional) Explore Netflow Collector and Analyzer Software

Netflow Collector and Analyzer Software is available from many vendors. An example is the Netflow Analyzer free edition by ManageEngine:

<https://www.manageengine.com/products/netflow/download-free.html>

Download and try it.

